FINN+: Project Overview

GitHub

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SAIL SUSTAINABLE LIFE-CYCLE OF INTELLIGENT SOCIO-TECHNICAL SYSTEMS

eki ENERGIEEFFIZIENTE KI DURCH DNN-APPROXIMATION FÜR FPGAs

PADERBORN UNIVERSITY

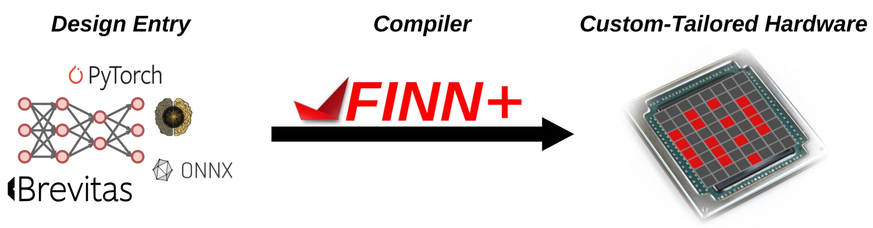
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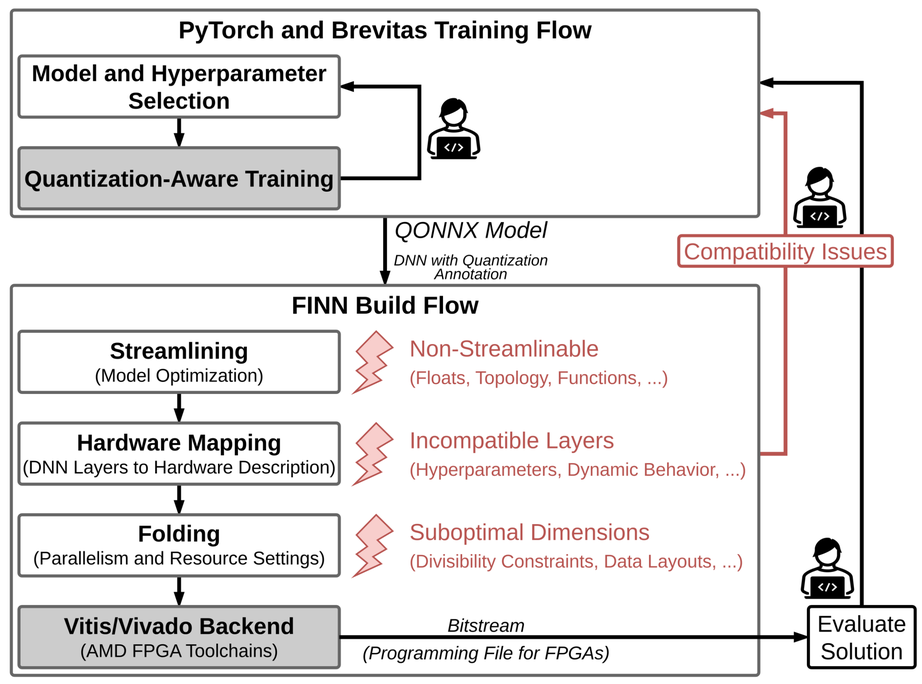
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Project Motivation

- **Co-design of DNN and inference accelerator architecture** offers unique performance and efficiency potential
- **Field Programmable Gate Arrays (FPGAs)** offer a re-configurable logic fabric that we harness to exploit such potential, e.g., by instantiating only the hardware that is needed
- **„Programming“ FPGAs is hard**, accelerator compilers are needed to explore this technology effectively, especially for non-domain experts



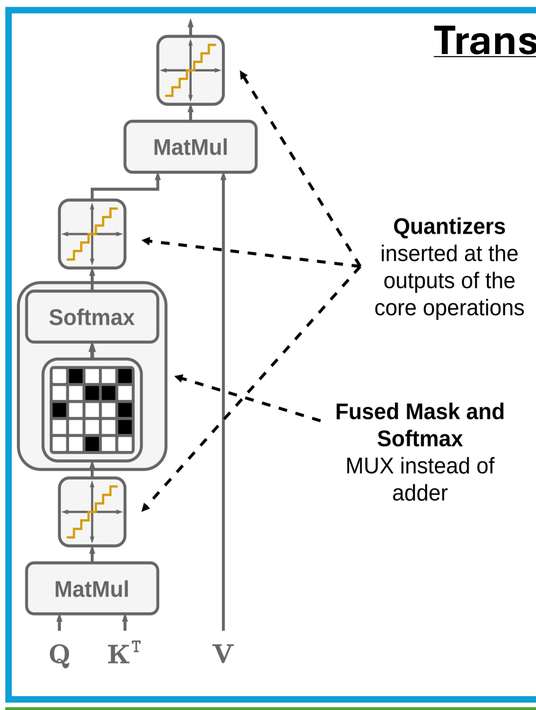
- **FINN+ is our fork of FINN:** a vehicle for advancing our research on dataflow compilers while aiming to improve **usability**, **flexibility**, **scalability**, and **automation**



Typical Co-Design Flow in traditional FINN, requiring significant user intervention

Core Contributions

Transformer Support



FINN-T: *Compiling Custom Dataflow Accelerators for Quantized Transformers*

- Enable advantages of streaming dataflow architectures for Transformers, i.e., small GPTs, BERTs, ViTs
- Flexibility and scaling behavior shown via systematic experiments and two case studies
- Identified challenges for future research which can be addressed with the new infrastructure

Re-imagined Streamlining

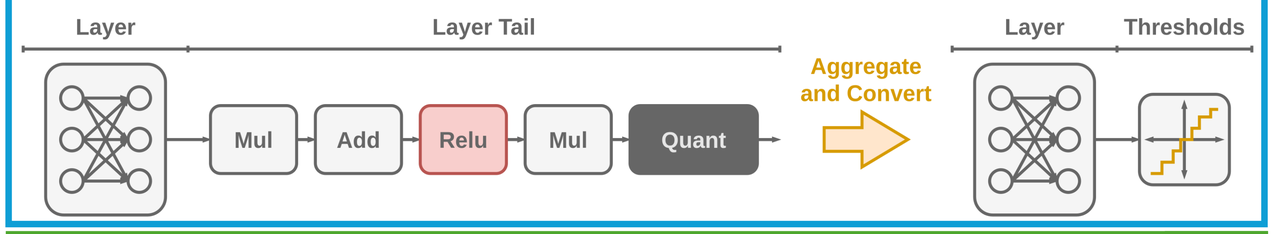
SIRA: *Scaled-Integer Range Analysis for Optimizing FPGA Dataflow Neural Network Accelerators*

“..., aggressive quantization can expose non-matrix-multiply operations as significant performance and resource bottlenecks on embedded systems.”

Complex Topologies
(e.g., ResNet, Transformer)

More activation functions
(e.g., Sigmoid, Silu)

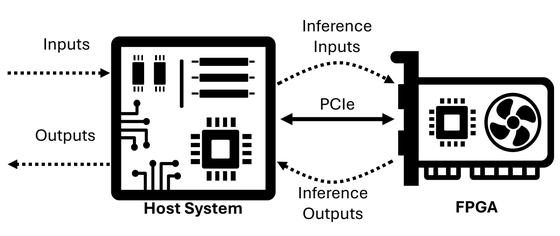
Float-Fallback
(binary-elementwise operations)

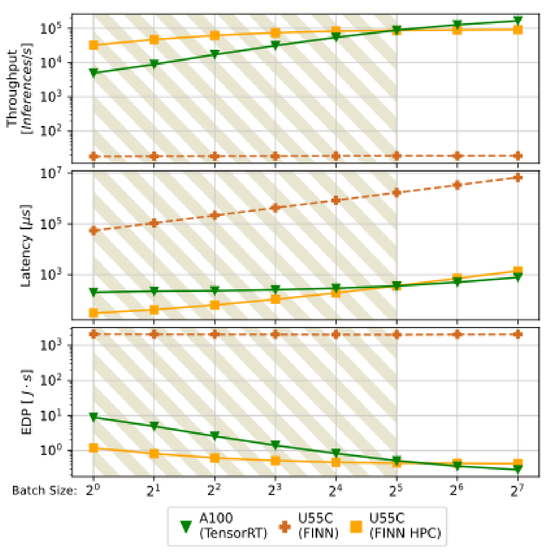


High-Performance Driver

FINN-HPC: *Closing the Gap for Energy-Efficient Neural Network Inference on FPGAs in HPC*

Optimized driver for performance/efficiency in datacenter deployments



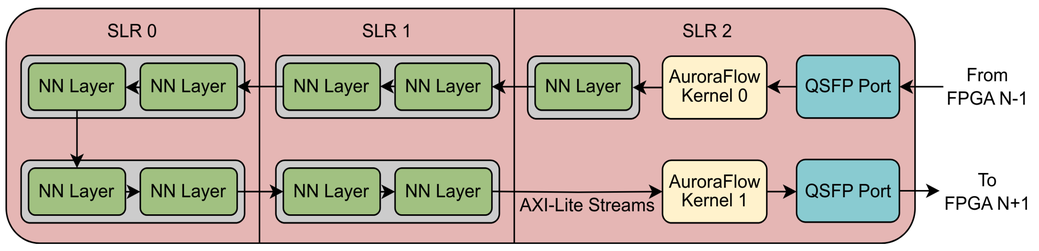


Throughput, Latency and Energy-Delay Product for a RadioML Transformer

Multi-FPGA Acceleration

AuroraFlow*, an Easy-to-Use, Low-Latency FPGA Communication Solution Demonstrated on Multi-FPGA Neural Network Inference*

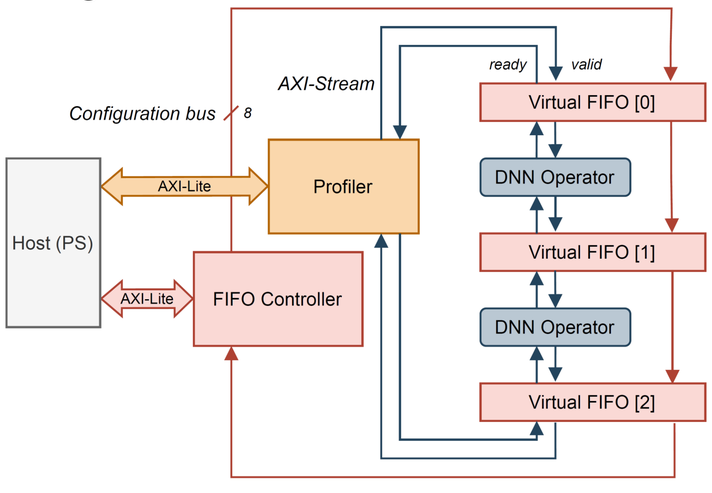
- Allows deployment of larger models and further unrolling of existing models to increase performanceby scaling accelerators to multiple FPGAs in a datacenter
- Communication via fiber for low latency, high throughput data transmission
- Integrated into FINN+ and configured via simple YAML configuration



Automated Buffer Sizing

LiveFIFO + Simulation-based FIFO Sizing
(see other posters)

- Dataflow accelerators require data buffering between layers
- We propose search-based methods to size these buffers optimally
- Sizing is performed either in **massively parallel simulation** or on an **FPGA-in-the-loop**
- Black-box approach supports complex topologies and requires no prior knowledge



Towards Hardware-aware AutoML

Empirical ***QoR Estimation*** *Flow for Fast Design Space Exploration of DNN Dataflow Accelerators*

- Micro-benchmarking of FINN operators (e.g. MatMul) for accurate resource & power measurement
- Data collection and fitting of ML models for resource/power prediction
- Speeds up (automated)design space exploration, e.g. via Neural Architecture Search

